

Session 1 — Linear algebra I

Exercise sheet

Work through these in order: they start with pure computation and become more conceptual toward the end. The recurring picture is a small data table — five participants, three measures — in which a row is one participant and a column is one measure. Show your reasoning; most answers are two or three lines. If a question defeats you, write “I don’t know” and we will work it out together.

The table used throughout (scores out of 10):

	RT	acc.	anx.
P_1	4	9	1
P_2	6	7	3
P_3	5	8	2
P_4	9	4	6
P_5	7	6	4

Exercise 1 One participant’s three scores form a list of three numbers — a vector in $\mathbb{R}^3$. Adding and scaling such vectors is all that “combining participants” can ever mean.

Let $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$ in $\mathbb{R}^2$, and $\mathbf{u} = (1, -2, 4)$, $\mathbf{v} = (0, 3, -1)$ in $\mathbb{R}^3$.

- (a) Compute $\mathbf{a} + \mathbf{b}$, $2\mathbf{a}$ and $\mathbf{a} - \mathbf{b}$.
- (b) Compute $\mathbf{u} + 2\mathbf{v}$ and $3\mathbf{u} - \mathbf{v}$.
- (c) Find λ, μ such that $\lambda(1, 1) + \mu(1, -1) = (3, 5)$.

Exercise 2 Using the table on the front page:

- (a) Write P_1 , P_2 and P_3 as vectors of $\mathbb{R}^3$, then check that $P_3 = \frac{1}{2}(P_1 + P_2)$.
- (b) Write the accuracy column as a vector. Which space does it live in?
- (c) In one sentence: why do a row and a column of the same table live in different vector spaces?

Exercise 3 Which of these families are linearly independent? For each dependent family, give an explicit combination of the vectors that vanishes.

- (a) $(1, 1)$ and $(2, 2)$;
- (b) $(1, 0)$ and $(0, 1)$;
- (c) $(2, -4)$ and $(-1, 2)$;
- (d) $(1, 2, 3)$ and $(2, 4, 7)$;

(e) $(0, 0)$ and $(1, 2)$.

Exercise 4 For which values of t are $(t, 2)$ and $(8, t)$ linearly dependent?

Exercise 5 Are $(1, 2)$, $(2, 4)$ and $(0, 1)$ independent in $\mathbb{R}^2$?

- (a) Answer first *without computing anything*, using only a dimension count.
 - (b) Now give an explicit vanishing combination.
 - (c) A questionnaire has 3 items. Can it produce more than 3 independent columns of data, however many people answer it?
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Exercise 6 *Anything you can build out of your measures by weighting and adding lies in their span. Nothing outside it is reachable, no matter how clever the weights.*

- (a) Is $(2, 5)$ in $\text{span}\{(1, 1), (1, -1)\}$? If so, give the coefficients.
 - (b) Describe $\text{span}\{(1, 1, 0), (0, 1, 1)\}$ inside $\mathbb{R}^3$: what shape is it, and is $(1, 2, 1)$ in it?
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Exercise 7 Compute the following dot products and lengths.

- (a) $(1, 2, 3) \cdot (4, 0, -1)$, $(2, -3) \cdot (4, 1)$ and $(1, 0, -2) \cdot (3, 5, 1)$.
 - (b) $\|(3, 4)\|$, $\|(6, 8)\|$, $\|(1, 1, 1)\|$ and $\|(2, -2, 1)\|$.
 - (c) The unit vector pointing along $(5, -12)$.
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Exercise 8 *A composite score — a scale total, an index, the prediction of a linear model — is a weighted sum of the measures. That is exactly a dot product between a weight vector $\mathbf{w}$ and a participant's profile $\mathbf{p}$.*

Take the weights $\mathbf{w} = (1, 2, -1)$: reward speed a little, accuracy a lot, penalise anxiety.

- (a) Compute the score of each of the five participants in the table.
 - (b) Who scores highest? Who scores lowest?
 - (c) Give a profile $\mathbf{p} \neq \mathbf{0}$ whose score is exactly 0.
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Exercise 9

- (a) Are $(1, 2)$ and $(2, -1)$ orthogonal? What does your answer mean for a scoring rule with weights $\mathbf{w} = (1, 2)$ applied to the profile $\mathbf{p} = (2, -1)$?

(b) For which k are $(k, 1)$ and $(k, -4)$ orthogonal?

Exercise 10

- (a) Compute $\cos \alpha$ for $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$.
- (b) Are $(1, 2)$ and $(2, 1)$ perpendicular? If not, give $\cos \alpha$.
- (c) Two participants have profiles $(2, 4)$ and $(4, 8)$. Compute their cosine similarity, and say in one sentence what the answer means and what it *ignores*.
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Exercise 11 Projection. In each case give both the signed shadow length and the projection vector.

- (a) Project $\mathbf{a} = (3, 1)$ onto the unit vector $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$.
- (b) Project $(1, 3)$ onto $\frac{1}{\sqrt{2}}(1, -1)$.
- (c) For (a), give the residual $\mathbf{a} - (\mathbf{a} \cdot \mathbf{u})\mathbf{u}$ and check it is orthogonal to $\mathbf{u}$.
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Exercise 12 Check the Cauchy–Schwarz inequality $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$ for $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (2, 4)$. Is it an equality? Why?

Exercise 13 *Among profiles of the same overall size, a scoring rule gives the highest score to the profile pointing the same way as its weights — and exactly zero to any profile perpendicular to them.*

A scoring rule has weights $\mathbf{w} = (1, 1)$. Among all profiles of unit length $\mathbf{p} = (\cos \theta, \sin \theta)$:

- (a) which one gives the largest score, and what is that score?
- (b) which ones give a score of zero?
- (c) In one sentence: what does it mean, practically, that a whole direction of variation can score zero?
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Exercise 14 Orthonormal families.

- (a) Is $\{(1, 0), (0, 1)\}$ orthonormal? Is $\{(1, 1), (1, -1)\}$?
- (b) Turn the second family into an orthonormal basis.
- (c) Complete $\frac{1}{\sqrt{2}}(1, -1)$ into an orthonormal basis of $\mathbb{R}^2$.
- (d) Give an orthonormal basis whose first axis points along $(3, 4)$.

Exercise 15 Work in the basis $\mathbf{e}_1 = \frac{1}{5}(3, 4)$, $\mathbf{e}_2 = \frac{1}{5}(-4, 3)$.

- (a) Give the coordinates of $(1, 0)$ in this basis.
- (b) Give the coordinates of $(5, 5)$, then rebuild $(5, 5)$ from them as a check.

Exercise 16 *Two similar tasks are usually correlated: someone good at A tends to be good at B. Recoding the pair as overall level and difference is a change of basis — and it is exactly what a contrast does.*

Take $\mathbf{f}_1 = \frac{1}{\sqrt{2}}(1, 1)$ (“overall level”) and $\mathbf{f}_2 = \frac{1}{\sqrt{2}}(-1, 1)$ (“B − A”).

- (a) Write $\mathbf{x} = (2, 0)$ in the basis $\{\mathbf{f}_1, \mathbf{f}_2\}$.
- (b) Give the coordinates of $(1, 3)$ in the same basis, and interpret the two numbers in words.
- (c) Verify on the previous item that $x_1^2 + x_2^2 = \|(1, 3)\|^2$.

Exercise 17 Rotate the axes of $\mathbb{R}^2$ by 30° .

- (a) Write down $\mathbf{e}_1$ and $\mathbf{e}_2$, then the coordinates of $(2, 0)$ in that basis.
- (b) Let $Q = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}$. Verify $Q^\top Q = I$.
- (c) In one sentence: why does a change to an orthonormal basis leave lengths unchanged?

Exercise 18 Starting from $(1, 1)$ and $(1, 0)$, produce an orthonormal basis of $\mathbb{R}^2$: take the first vector as the first axis, subtract from the second vector its component along that axis, then normalise. (*This is Gram–Schmidt in two dimensions.*)

Exercise 19 A matrix is a machine that eats a vector and returns a vector. In each case compute the output and say in words what the matrix does.

- (a) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
- (b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 5 \\ -2 \end{pmatrix}$.
- (c) Which 2×2 matrix rotates the plane by 90° anticlockwise? Check it on $(1, 0)$.

Exercise 20 *Recoding a pair of task scores as mean and difference is one matrix applied to every participant.*

Let $M = \begin{pmatrix} 1/2 & 1/2 \\ -1 & 1 \end{pmatrix}$ and $\mathbf{s} = (8, 4)$.

- (a) Compute $M\mathbf{s}$ twice: once by rows, once as a combination of the columns.
 - (b) Read the two output numbers in words.
 - (c) Is M an orthonormal change of basis? Justify in one line.
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Exercise 21 *Collect the five participants as the rows of a 5×3 matrix X . Then $X\mathbf{w}$ applies one scoring rule to everybody in a single product — which is the linear part of any regression prediction.*

Let X be the table on the front page and $\mathbf{w} = (1, 2, -1)$ as in Exercise 8.

- (a) What are the dimensions of X , of $\mathbf{w}$, and of $X\mathbf{w}$?
 - (b) Compute $X\mathbf{w}$ and check it against your answers to Exercise 8(a).
 - (c) What does the entry X_{42} represent? And X_{24} ?
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Exercise 22 Matrix products.

- (a) Compute $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -1 \end{pmatrix}$ twice: once by rows, once as a combination of the columns.
 - (b) Compute $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ and the same product in the opposite order. Comment.
 - (c) Compute $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$ and, again, the opposite order.
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Exercise 23 Transposes.

- (a) Write the transpose of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and of $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$.
- (b) With $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, verify that $(AB)^\top = B^\top A^\top$.
- (c) If X is the 5×3 table of Exercise 21, what are the dimensions of $X^\top X$ and of $XX^\top$? (We will meet the first one again next session.)